

LECTURE 13 (153-248)

$$y_t = \beta_0 + \beta_1(t-1) + \sum_{j=2}^{n-1} \beta_j \text{ReLU}(t-j) + \varepsilon_t$$

$$y = X\beta + \varepsilon$$

$$X_{n \times n} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$$

$\hat{\beta}_{\text{ridge}}(\lambda)$ minimizes

$$\|y - X\beta\|^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2$$

$$\hat{\beta}_{\text{ridge}}(\lambda) = (X^T X + \lambda J)^{-1} X^T y$$

$$J_{n \times n} = \begin{bmatrix} 0 & 0 & & 0 \\ 0 & 1 & & \\ & 0 & \ddots & \\ 0 & & & 1 \end{bmatrix}$$

$$L(\beta) = \|y - X\beta\|^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2$$

$$= (y - X\beta)^T (y - X\beta) + \lambda \beta^T J \beta$$

$$= y^T y - 2y^T X\beta + \underbrace{\beta^T X^T X \beta + \lambda \beta^T J \beta}$$

$$\begin{aligned}
 &= \beta^T (x^T x + \lambda J) \beta - 2y^T x \beta + y^T y \\
 &= \left(\beta - (x^T x + \lambda J)^{-1} x^T y \right)^T (x^T x + \lambda J) \beta \\
 &\quad + y^T y - y^T x (x^T x + \lambda J)^{-1} x^T y
 \end{aligned}$$

$$x^2 - 7x + 10$$

$$(x - 3.5)^2 + 10 - 3.5^2$$

Formula: $L(\beta) = \|y - X\beta\|^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2$

$$\begin{aligned}
 &= \left(\beta - \hat{\beta}_{\text{ridge}}(\lambda) \right)^T (x^T x + \lambda J) \left(\beta - \hat{\beta}_{\text{ridge}}(\lambda) \right) \\
 &\quad + y^T y - y^T x (x^T x + \lambda J)^{-1} x^T y \\
 &\Rightarrow \text{minimizer is } \hat{\beta}_{\text{ridge}}(\lambda) = (x^T x + \lambda J)^{-1} x^T y
 \end{aligned}$$

BAYESIAN REGULARIZATION

Ridge regression "corresponds" to the following prior:

$$\beta_0, \beta_1 \sim \text{Unif}(-C, C) \text{ or } N(0, C), \quad \beta_2, \dots, \beta_{n-1} \sim N\left(0, \frac{\sigma^2}{\lambda}\right)$$

$$f(\beta_0, \beta_1, \dots, \beta_{n-1}) \propto \prod_{j=2}^{n-1} \frac{1}{\sqrt{\sigma^2/\lambda}} \exp\left(-\frac{\lambda \beta_j^2}{2\sigma^2}\right)$$

$$= \frac{\lambda^{\frac{n-2}{2}} \sigma^{-(n-2)}}{\lambda^{\frac{n-2}{2}} \sigma^{-(n-2)}} \exp\left(-\frac{\lambda \sum_{j=2}^{n-1} \beta_j^2}{2\sigma^2}\right)$$

posterior of $\beta \propto$ prior $\times$ likelihood

$$\propto \frac{\lambda^{\frac{n-2}{2}} \sigma^{-(n-2)}}{\lambda^{\frac{n-2}{2}} \sigma^{-(n-2)}} \exp\left(-\frac{\lambda \sum_{j=2}^{n-1} \beta_j^2}{2\sigma^2}\right) \underbrace{\frac{1}{\sigma^n} \exp\left(-\frac{\|y - X\beta\|^2}{2\sigma^2}\right)}_{\text{likelihood}}$$

$$= \frac{\lambda^{\frac{n-2}{2}} \sigma^{-2n+2}}{\lambda^{\frac{n-2}{2}} \sigma^{-2n+2}} \exp\left(-\frac{\|y - X\beta\|^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2}{2\sigma^2}\right)$$

$$= \frac{\lambda^{\frac{n-2}{2}} \sigma^{-2n+2}}{\lambda^{\frac{n-2}{2}} \sigma^{-2n+2}} \exp\left(-\frac{L(\beta)}{2\sigma^2}\right)$$

$$= \frac{\lambda^{\frac{n-2}{2}} \sigma^{-2n+2}}{\lambda^{\frac{n-2}{2}} \sigma^{-2n+2}} \exp\left(-\frac{(\beta - \hat{\beta}_{\text{ridge}}(\lambda))^T (X^T X + \lambda I) (\beta - \hat{\beta}_{\text{ridge}}(\lambda))}{2\sigma^2}\right)$$

$$\exp\left[-\frac{y^T y - y^T X (X^T X + \lambda I)^{-1} X^T y}{2\sigma^2}\right]$$

$$\Rightarrow \beta \mid \text{data}, \sigma, \lambda \sim N\left(\hat{\beta}_{\text{ridge}}(\lambda), \sigma^2 (X^T X + \lambda I)^{-1}\right)$$

$$\exp \left[- \frac{(\beta - \hat{\beta}_{\text{ridge}})^T (X^T X + \lambda J) (\beta - \hat{\beta}_{\text{ridge}})}{2\sigma^2} \right]$$

$$\exp \left[- \frac{1}{2} (\beta - \mu)^T \Sigma^{-1} (\beta - \mu) \right]$$

$$\mu = \hat{\beta}_{\text{ridge}} \quad \Sigma^{-1} = \frac{X^T X + \lambda J}{\sigma^2}$$

$$\Rightarrow \Sigma = \sigma^2 (X^T X + \lambda J)^{-1}$$

RIDGE REGRESSION

= POSTERIOR MEAN of

Prior $\beta_0, \beta_1 \sim \text{Unif}(-C, C)$, $\beta_2, \dots, \beta_{n-1} \stackrel{\text{iid}}{\sim} N(0, \frac{\sigma^2}{\lambda})$
 Likelihood $y \sim N(X\beta, \sigma^2 I)$

Full Bayesian Model

$\beta_0, \beta_1 \sim \text{Unif}(-C, C)$, $\beta_2, \dots, \beta_{n-1} \stackrel{\text{iid}}{\sim} N(0, \frac{\sigma^2}{\lambda})$

$\log \sigma \sim \text{Unif}(-C, C)$, $\lambda \sim f(\lambda)$
 $\downarrow \lambda$
 prior for λ .

$\log \lambda \sim \text{Unif}(\log \lambda_{\min}, \log \lambda_{\max})$

Fitted curve:

$$\beta_0 + \beta_1 (t-1) + \beta_2 (t-2) + \dots + \beta_{n-1} (t-(n-1))$$

$$f(\beta, \sigma, \lambda) \propto f(\beta, \sigma, \lambda) \times \text{likelihood}$$

$$\frac{f(\beta, \sigma, \lambda)}{\beta, \sigma, \lambda | \text{data}} \propto \lambda^{\frac{n-2}{2}} \sigma^{-2n+2} \exp \left[- \frac{(\beta - \hat{\beta}_{\text{ridge}})^T (X^T X + \lambda I) (\beta - \hat{\beta}_{\text{ridge}})}{2\sigma^2} \right]$$

$$\exp \left[- \frac{y^T y - y^T X (X^T X + \lambda I)^{-1} X^T y}{2\sigma^2} \right]$$

$$\times \frac{1}{\sigma} \mathbb{I}(\bar{e}^c < \sigma < e^c) f(\lambda)$$

$$\propto f(\lambda) \lambda^{\frac{n-2}{2}} \sigma^{-2n+1} \exp \left(- \frac{(\beta - \hat{\beta}_{\text{ridge}})^T (X^T X + \lambda I) (\beta - \hat{\beta}_{\text{ridge}})}{2\sigma^2} \right)$$

$$\mathbb{I}(\sigma > 0) \exp \left[- \frac{y^T y - y^T X (X^T X + \lambda I)^{-1} X^T y}{2\sigma^2} \right]$$

$$\beta | \text{data}, \lambda, \sigma \sim N(\hat{\beta}_{\text{ridge}}, \sigma^2 (X^T X + \lambda I)^{-1})$$

$$\int \exp \left(- \frac{(\beta - \hat{\beta}_{\text{ridge}})^T (X^T X + \lambda I) (\beta - \hat{\beta}_{\text{ridge}})}{2\sigma^2} \right) d\beta$$

$$\propto \sqrt{\det(\sigma^2 (X^T X + \lambda I)^{-1})} = \sigma^n |X^T X + \lambda I|^{-1/2}$$

$$f(\sigma, \lambda) \propto f(\sigma, \lambda) \times \text{likelihood}$$

$$|A| = \det(A)$$

$$\propto f_{\lambda} \propto \lambda^{\frac{n-2}{2}} \sigma^{-n+1} \exp\left[-\frac{y^T y - y^T X(X^T X + \lambda I)^{-1} X^T y}{2\sigma^2}\right]$$

$I(\sigma > 0)$

Now integrate over σ : LEADS TO

$$\frac{\sigma}{\sqrt{y^T y - y^T X(X^T X + \lambda I)^{-1} X^T y}} = \lambda$$

$$f_{\lambda} \propto \lambda^{\frac{n-2}{2}} |X^T X + \lambda I|^{-1/2} \frac{1}{[y^T y - y^T X(X^T X + \lambda I)^{-1} X^T y]^{\frac{n}{2}-1}}$$

$$f_{\lambda} \propto f_{\lambda} f_{\text{data}|\lambda}$$

$$f_{\text{data}|\lambda} \propto \lambda^{\frac{n-2}{2}} |X^T X + \lambda I|^{-1/2} \frac{1}{[y^T y - y^T X(X^T X + \lambda I)^{-1} X^T y]^{\frac{n}{2}-1}}$$

Likelihood:

$$f_{\text{data}|\beta, \sigma} \propto \frac{1}{\sigma^n} \exp\left(-\frac{\|y - X\beta\|^2}{2\sigma^2}\right)$$

$$f_{\text{data}|\lambda} = \iint f_{\text{data}|\beta, \sigma, \lambda} f_{\beta, \sigma|\lambda} d\beta d\sigma$$

$$\beta_0, \beta_1 \sim \text{Unif}(-C, C)$$

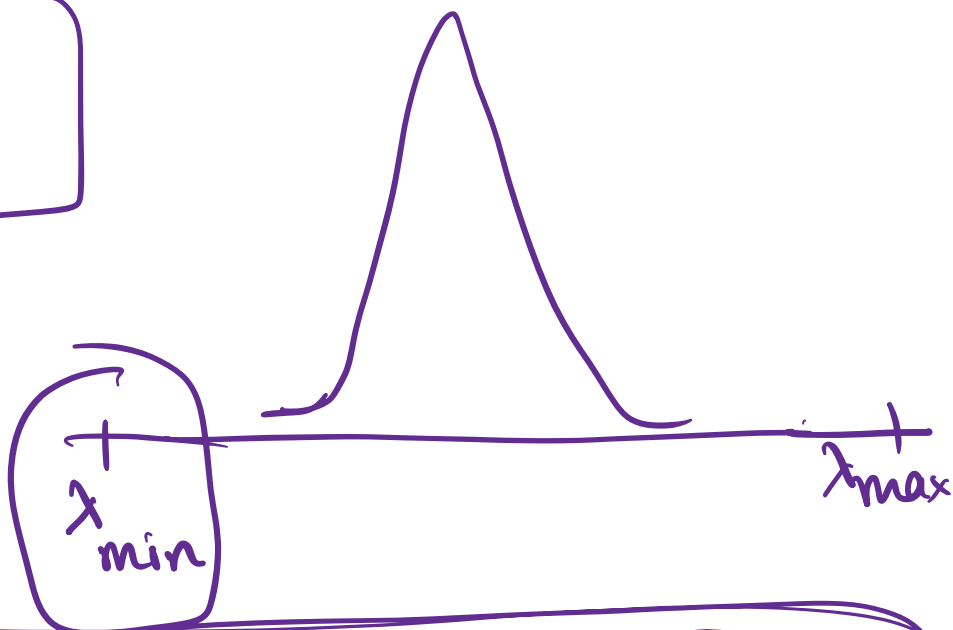
$$\beta_2 \dots \beta_{n-1} \sim N(0, \frac{\sigma^2}{\lambda})$$

$$\log \sigma \sim \text{Unif}(-C, C)$$

$$\min \|y - X\beta\|$$

Integrated or Marginal Likelihood

$$f_{\text{data}}(\lambda)$$



$$f_{\text{data}}(\lambda) = \int f_{\text{data}}(\beta) f_{\beta|\lambda}(\beta) d\beta$$

takes high values when β is the overfitted solution

$$\beta \sim N(0, \frac{\sigma^2}{\lambda})$$